

Coherent-Constructible Correspondence

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1.1 Standard Notation

The toric notation used in this article will be the standard one. We will not reintroduce the full definition of a toric variety each time it appears. Instead, for a toric variety X , we always regard it as equipped with its acting algebraic torus

$$T \cong (\mathbb{C}^*)^n$$

and with the usual dual lattices

$$N = \text{Hom}(\mathbb{C}^*, T), \quad M = \text{Hom}(T, \mathbb{C}^*).$$

The lattice N is the lattice of one-parameter subgroups, while M is the character lattice. They satisfy

$$M = \text{Hom}_{\mathbb{Z}}(N, \mathbb{Z}).$$

We denote their realifications by

$$N_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R}, \quad M_{\mathbb{R}} = M \otimes_{\mathbb{Z}} \mathbb{R}.$$

In this article, our main theorem is the an equivalence of stable ∞ -category

$$\kappa : \mathrm{Perf}_T(X_\Sigma) \xrightarrow{\sim} \mathrm{Sh}_{cc}(M_{\mathbb{R}}, \Lambda_\Sigma)$$

2 Symplectic Aspects of Toric Varieties

Let X be a smooth projective toric variety. Since X is projective, its underlying complex manifold admits a Kähler form. In particular, X may be viewed as a symplectic manifold. The purpose of this section is to recall the basic symplectic notions needed for this perspective and to explain, at the level of definitions, how the symplectic structure on X is obtained from the algebraic data of an ample toric line bundle.

We shall also fix notation and introduce several related concepts that will be used later in the discussion of toric coherent-constructible correspondence. To keep the exposition concise, we will state the standard facts used in this section without proof.

In this chapter, we will use both the Zariski topology and the usual analytic topology when needed, and we will regard X both as an algebraic variety and as its associated complex manifold; when no confusion can arise, we will not explicitly distinguish between these viewpoints or topologies.

2.1 Symplectic Manifolds

Let X be a smooth real manifold. A **symplectic form** on X is a closed non-degenerate two-form

$$\omega \in \Omega^2(X),$$

that is,

$$d\omega = 0,$$

and for every point $x \in X$, the bilinear form

$$\omega_x : T_x X \times T_x X \rightarrow \mathbb{R}$$

is non-degenerate. Equivalently, the map

$$T_x X \longrightarrow T_x^* X, \quad v \longmapsto \omega_x(v, -)$$

is an isomorphism for every $x \in X$. A pair (X, ω) is called a **symplectic manifold**.

A diffeomorphism

$$\phi : X \rightarrow X$$

is called a **symplectomorphism** if it preserves the symplectic form:

$$\phi^* \omega = \omega.$$

Thus a symplectomorphism is an automorphism of the symplectic manifold (X, ω) .

Let G be a Lie group acting smoothly on X . We say that the action is a **symplectic action** if every element $g \in G$ acts by a symplectomorphism, that is,

$$g^* \omega = \omega$$

for all $g \in G$. We denote the action of G by

$$\alpha : G \times X \rightarrow X, \quad (g, x) \mapsto g \cdot x.$$

For each point $x \in X$, the action determines an orbit map

$$\alpha_x : G \rightarrow X, \quad g \mapsto g \cdot x.$$

Let $\mathfrak{g}$ be the Lie algebra of G . For $\xi \in \mathfrak{g}$, the corresponding one-parameter subgroup is

$$t \mapsto \exp(t\xi).$$

Differentiating the orbit map along this one-parameter subgroup gives a tangent vector at x :

$$\xi_X(x) = \left. \frac{d}{dt} \right|_{t=0} \exp(t\xi) \cdot x \in T_x X.$$

As x varies, this defines a vector field

$$\xi_X \in \Gamma(TX).$$

The assignment

$$\mathfrak{g} \rightarrow \Gamma(TX), \quad \xi \mapsto \xi_X$$

is called the **infinitesimal action** of G on X . The vector field ξ_X is called the **fundamental vector field** generated by ξ .

A symplectic G -action is called **Hamiltonian** if there exists a smooth map

$$\mu : X \rightarrow \mathfrak{g}^*$$

such that, for every $\xi \in \mathfrak{g}$,

$$d\langle \mu, \xi \rangle = \iota_{\xi_X} \omega.$$

Here

$$\langle \mu, \xi \rangle : X \rightarrow \mathbb{R}$$

denotes the function obtained by pairing $\mu(x) \in \mathfrak{g}^*$ with $\xi \in \mathfrak{g}$, and $\iota_{\xi_X} \omega$ denotes contraction of ω with the vector field ξ_X . The map

$$\mu : X \rightarrow \mathfrak{g}^*$$

is called a **moment map**. The triple (X, ω, μ) is called a **Hamiltonian G -space**.

The moment map is not unique in general. If $\mu : X \rightarrow \mathfrak{g}^*$ is a moment map, then $\mu + c$ is also a moment map for any constant element $c \in \mathfrak{g}^*$. Thus a moment map is determined only up to addition of a constant element of $\mathfrak{g}^*$.

A moment map is called **G -equivariant** if

$$\mu(g \cdot x) = \text{Ad}_g^* \mu(x)$$

for all $g \in G$ and $x \in X$, where Ad^* is the coadjoint action, namely the dual action to the adjoint representation

$$\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g}).$$

2.2 Kähler Manifolds and Complex Structure

Let X be a smooth manifold, and let $D \subset TX$ be a smooth subbundle of the tangent bundle. We call D a **distribution** on X .

The distribution D is called **integrable** if for every point $x \in X$, there exists an immersed submanifold $Y \subset X$ passing through x such that

$$T_y Y = D_y$$

for every $y \in Y$. Such a submanifold Y is called an **integral manifold** of D .

An **almost complex structure** on X is a bundle endomorphism

$$J : TX \rightarrow TX$$

such that

$$J^2 = -\text{id}_{TX}.$$

If J is integrable, then X is a complex manifold, and J is called its **complex structure**.

Suppose now that X is a complex manifold with complex structure J .

A Riemannian metric g on X is called **Hermitian** if it is compatible with J , namely

$$g(Jv, Jw) = g(v, w)$$

for all tangent vectors v, w . Associated to such a Hermitian metric is a real two-form

$$\omega(v, w) = g(Jv, w).$$

This two-form is called the **fundamental form** of the Hermitian metric.

The Hermitian metric g is called **Kähler** if

$$d\omega = 0.$$

In this case ω is called a **Kähler form**, and the triple

$$(X, J, \omega)$$

is called a **Kähler manifold**.

A Kähler form is in particular a symplectic form. Indeed, it is a closed non-degenerate real two-form. Thus every Kähler manifold is naturally a symplectic manifold. Conversely, on a complex manifold, a symplectic form compatible with the complex structure determines a Kähler metric by

$$g(v, w) = \omega(v, Jw).$$

2.3 Hermitian Line Bundles and Curvature

We recall a standard construction from complex differential geometry. Let X be a complex manifold, and let L be a holomorphic line bundle. A **Hermitian metric** on L is a smoothly varying Hermitian inner product

$$h_x : L_x \times L_x \rightarrow \mathbb{C}$$

on the fibers of L . If s is a local non-vanishing holomorphic frame of L , then the metric is determined locally by the positive smooth function

$$h(s, s) > 0.$$

There is a unique connection ∇_h on L , called the **Chern connection**, which is compatible with both the holomorphic structure and the Hermitian metric. The compatibility with the holomorphic structure means that

$$\nabla_h^{0,1} = \bar{\partial}_L,$$

while compatibility with the metric means that

$$dh(s_1, s_2) = h(\nabla_h s_1, s_2) + h(s_1, \nabla_h s_2)$$

for smooth local sections s_1, s_2 of L . The **curvature** of the Chern connection is

$$F_h = \nabla_h^2.$$

Since L is a line bundle, F_h may be regarded as a complex-valued two-form on X . Locally, if s is a holomorphic frame and

$$h(s, s) = e^{-\varphi},$$

then the curvature is expressed in terms of the local potential φ . With the usual convention, the associated real $(1, 1)$ -form is

$$\omega_h = \sqrt{-1} \partial \bar{\partial} \varphi.$$

Equivalently, up to the chosen sign convention, one may write

$$\omega_h = \sqrt{-1} F_h.$$

If L is positive, then ω_h is a Kähler form. Thus a positive Hermitian metric on a positive holomorphic line bundle produces a symplectic form on the underlying smooth manifold.

2.4 Algebraic Tori and Compact Real Tori

Let X be a smooth projective toric variety of complex dimension n . The algebraic torus T is a complex algebraic group. It contains a distinguished compact real Lie subgroup, namely its maximal compact real subtorus

$$T_{\mathbb{R}} \cong (S^1)^n.$$

Concretely, under the identification $T \cong (\mathbb{C}^*)^n$, this subgroup is

$$T_{\mathbb{R}} = \{(z_1, \dots, z_n) \in (\mathbb{C}^*)^n \mid |z_1| = \dots = |z_n| = 1\}.$$

The **compact torus action**

$$T_{\mathbb{R}} \times X \longrightarrow X.$$

is obtained by restricting the algebraic T -action on X to the subgroup $T_{\mathbb{R}} \subset T$.

The real vector space $N_{\mathbb{R}}$ is naturally identified with the Lie algebra of the compact torus $T_{\mathbb{R}}$:

$$N_{\mathbb{R}} \cong \text{Lie}(T_{\mathbb{R}}).$$

Dually, we have

$$M_{\mathbb{R}} \cong \text{Lie}(T_{\mathbb{R}})^*.$$

Under these identifications, the **exponential map of the compact torus** is

$$\exp : T_{\mathbb{R}}^{\text{Lie}} \cong N_{\mathbb{R}} \longrightarrow T_{\mathbb{R}},$$

and its kernel is the lattice N . Hence

$$T_{\mathbb{R}} \cong N_{\mathbb{R}}/N.$$

Similarly, the **dual compact torus** is

$$T_{\mathbb{R}}^{\vee} \cong M_{\mathbb{R}}/M.$$

2.5 Kähler Structures from Ample Toric Line Bundles

Let X be a smooth projective toric variety over $\mathbb{C}$, and let

$$L = \mathcal{O}_X(D)$$

be an ample T -linearized line bundle on X . Passing to the analytification, we regard L as a holomorphic line bundle over the complex manifold X . Since L is ample, one can choose a positive smooth Hermitian metric h on L . Let ∇_h be the associated Chern connection and let

$$F_h = \nabla_h^2$$

be its curvature. With a fixed sign convention, the form

$$\omega_D = \sqrt{-1}F_h$$

is a Kähler form on X . In particular, ω_D is a closed non-degenerate real two-form, so

$$(X, \omega_D)$$

is a symplectic manifold.

Moreover, since the compact real torus $T_{\mathbb{R}}$ is compact, we may average the Hermitian metric h over $T_{\mathbb{R}}$. Thus h , and hence ω_D , may be chosen to be $T_{\mathbb{R}}$ -invariant. Therefore the algebraic data $(X, \mathcal{O}_X(D))$ gives rise to a $T_{\mathbb{R}}$ -invariant Kähler, hence symplectic, manifold (X, ω_D) .

Since ω_D may be chosen to be $T_{\mathbb{R}}$ -invariant, the compact torus action on X_{Σ} is symplectic. Using the identifications

$$\mathrm{Lie}(T_{\mathbb{R}}) \cong N_{\mathbb{R}}, \quad \mathrm{Lie}(T_{\mathbb{R}})^* \cong M_{\mathbb{R}},$$

a moment map for this action is a smooth map

$$\mu_D : X_{\Sigma} \rightarrow M_{\mathbb{R}}$$

such that

$$d\langle \mu_D, \xi \rangle = \iota_{\xi_X} \omega_D$$

for every $\xi \in N_{\mathbb{R}}$. Thus the pair $(X_{\Sigma}, \mathcal{O}_X(D))$ produces a Hamiltonian $T_{\mathbb{R}}$ -space

$$(X_{\Sigma}, \omega_D, \mu_D).$$

The image of μ_D will be discussed in the next section.

2.6 Moment Polytope

Let X be a smooth projective toric variety. Let

$$D = \sum_{\rho \in \Sigma(1)} c_{\rho} D_{\rho}$$

be a T -invariant Cartier divisor, where D_{ρ} is the torus-invariant divisor corresponding to the ray

$$\rho = \mathbb{R}_{\geq 0} v_{\rho}$$

and $v_{\rho} \in N$ is the primitive generator of ρ . Assume that

$$L = \mathcal{O}_X(D)$$

is ample.

Definition 2.1 (Moment polytope). The polytope associated to D is

$$\Delta_D = \{m \in M_{\mathbb{R}} \mid \langle m, v_\rho \rangle \geq -c_\rho \text{ for all } \rho \in \Sigma(1)\}.$$

The main fact we need is the following.

Theorem 2.2. With the standard normalization determined by the divisor D , the image of the moment map

$$\mu_D : X_\Sigma \rightarrow M_{\mathbb{R}}$$

associated to the ample line bundle $\mathcal{O}_X(D)$ is precisely the polytope Δ_D :

$$\mu_D(X_\Sigma) = \Delta_D.$$

Proof. We use the projective embedding determined by the complete linear system of $L = \mathcal{O}_X(D)$. By the standard description of global sections of toric line bundles, we have

$$H^0(X, \mathcal{O}_X(D)) = \bigoplus_{m \in \Delta_D \cap M} \mathbb{C} \cdot \chi^m.$$

Thus the lattice points of Δ_D give a basis of global sections. Writing

$$\Delta_D \cap M = \{m_0, \dots, m_s\},$$

the complete linear system defines a T -equivariant projective embedding

$$\Phi_D : X_\Sigma \longrightarrow \mathbb{P}^s.$$

On the dense torus $T \subset X_\Sigma$, this map is given by

$$t \longmapsto [\chi^{m_0}(t) : \dots : \chi^{m_s}(t)].$$

We endow $\mathbb{P}^s$ with the Fubini–Study Kähler form and pull it back to X_Σ . The corresponding moment map, restricted to the dense torus, is

$$\mu_D(t) = \frac{\sum_{m \in \Delta_D \cap M} m |\chi^m(t)|^2}{\sum_{m \in \Delta_D \cap M} |\chi^m(t)|^2}.$$

Therefore $\mu_D(t)$ is a weighted average of the lattice points $m \in \Delta_D \cap M$, with positive real weights

$$|\chi^m(t)|^2.$$

Hence

$$\mu_D(T) \subset \text{Conv}(\Delta_D \cap M) = \Delta_D.$$

Since T is dense in X_Σ and μ_D is continuous, it follows that

$$\mu_D(X_\Sigma) \subset \Delta_D.$$

It remains to see that every point of Δ_D occurs in the image. On the dense torus, write

$$t = \exp(y + i\theta), \quad y \in N_{\mathbb{R}}, \quad \theta \in N_{\mathbb{R}}/N.$$

Then

$$|\chi^m(t)|^2 = e^{2\langle m, y \rangle}.$$

Thus the moment map on the open orbit can be written as

$$\mu_D(y) = \frac{\sum_{m \in \Delta_D \cap M} m e^{2\langle m, y \rangle}}{\sum_{m \in \Delta_D \cap M} e^{2\langle m, y \rangle}}.$$

Equivalently,

$$\mu_D(y) = \frac{1}{2} \nabla_y \log \left(\sum_{m \in \Delta_D \cap M} e^{2\langle m, y \rangle} \right).$$

The gradient of the strictly convex function

$$y \mapsto \log \left(\sum_{m \in \Delta_D \cap M} e^{2\langle m, y \rangle} \right)$$

maps $N_{\mathbb{R}}$ onto the interior of the convex hull of the weights $\Delta_D \cap M$, namely onto Δ_D° . Hence

$$\Delta_D^\circ \subset \mu_D(T).$$

Taking closures gives

$$\Delta_D \subset \mu_D(X_\Sigma).$$

Combining the two inclusions, we obtain

$$\mu_D(X_\Sigma) = \Delta_D.$$

□

Thus the algebraic datum of the ample toric line bundle $\mathcal{O}_X(D)$ determines not only a Kähler form ω_D , but also a canonical convex polytope Δ_D , namely the image of the associated moment map.

3 Categorical and Sheaf-Theoretic Preliminaries

The coherent–constructible correspondence is an equivalence between two derived categories of sheaves. On the coherent side one considers equivariant coherent sheaves on a toric variety, while on the constructible side one considers constructible sheaves whose singular support is constrained by a conic Lagrangian subset determined by the fan. Since morphisms in these categories are naturally derived morphisms, it is not sufficient to regard them merely as ordinary categories. We will therefore use the language of categories enriched over complexes, or equivalently dg categories, in order to keep track of Hom-complexes, shifts, cones, and derived equivalences.

The purpose of this section is only to fix the categorical conventions needed later. We recall the notion of an enriched category, specialize to dg categories, and explain how their homotopy categories recover the triangulated categories appearing in the statement of the CCC. We then review the sheaf-theoretic categories used on both sides of the correspondence: coherent and equivariant coherent sheaves, constructible sheaves, standard and costandard sheaves, singular support, and sheaves with prescribed singular support.

3.1 dg Categories

We first recall the notion of an enriched category. In an ordinary category, the morphisms between two objects form a set. In many geometric situations, however, the morphisms carry additional structure. For example, in derived categories the morphisms are naturally organized into complexes, whose cohomology groups give the corresponding Ext groups. This motivates the notion of a category enriched over a monoidal category.

A **monoidal category** is a category $\mathcal{V}$ together with a bifunctor

$$\otimes : \mathcal{V} \times \mathcal{V} \longrightarrow \mathcal{V},$$

an object $\mathbf{1} \in \mathcal{V}$, called the unit object, and natural isomorphisms

$$(X \otimes Y) \otimes Z \simeq X \otimes (Y \otimes Z), \quad \mathbf{1} \otimes X \simeq X, \quad X \otimes \mathbf{1} \simeq X,$$

satisfying pentagon axiom

$$\begin{array}{ccc} ((W \otimes X) \otimes Y) \otimes Z & \xrightarrow{\alpha_{W \otimes X, Y, Z}} & (W \otimes X) \otimes (Y \otimes Z) \xrightarrow{\alpha_{W, X, Y \otimes Z}} W \otimes (X \otimes (Y \otimes Z)) \\ \alpha_{W, X, Y} \otimes \text{id}_Z \downarrow & & \nearrow \text{id}_W \otimes \alpha_{X, Y, Z} \\ (W \otimes (X \otimes Y)) \otimes Z & \xrightarrow{\alpha_{W, X \otimes Y, Z}} & W \otimes ((X \otimes Y) \otimes Z) \end{array}$$

and triangle axiom

$$\begin{array}{ccc} (X \otimes \mathbf{1}) \otimes Y & \xrightarrow{\alpha_{X, \mathbf{1}, Y}} & X \otimes (\mathbf{1} \otimes Y) \\ \rho_X \otimes \text{id}_Y \searrow & & \swarrow \text{id}_X \otimes \lambda_Y \\ & X \otimes Y & \end{array}$$

The bifunctor $\otimes$ is called the **tensor product**.

The basic examples of monoidal categories are **Set** with the cartesian product, **Ab** with the tensor product of abelian groups, and $\text{Ch}(k)$ with the tensor product of complexes of k -modules. In the last example, the unit object is the complex k concentrated in degree zero.

A **category enriched in $\mathcal{V}$** , or a **$\mathcal{V}$ -enriched category**, is a category-like object whose morphisms are not sets, but objects of $\mathcal{V}$. More precisely, a $\mathcal{V}$ -enriched category $\mathcal{C}$ consists of:

1. a collection of objects $\text{Ob}(\mathcal{C})$;
2. for every pair of objects $A, B \in \mathcal{C}$, an object

$$\text{Hom}_{\mathcal{C}}(A, B) \in \mathcal{V};$$

3. for every triple $A, B, C \in \mathcal{C}$, a composition morphism in $\mathcal{V}$

$$\text{Hom}_{\mathcal{C}}(B, C) \otimes \text{Hom}_{\mathcal{C}}(A, B) \longrightarrow \text{Hom}_{\mathcal{C}}(A, C);$$

4. for every object $A \in \mathcal{C}$, a unit morphism

$$\mathbf{1} \longrightarrow \text{Hom}_{\mathcal{C}}(A, A).$$

These data are required to satisfy the usual associativity and unit axioms, written using the monoidal structure of $\mathcal{V}$.

Ordinary categories are recovered by taking

$$\mathcal{V} = \text{Set}$$

with its Cartesian monoidal structure. In this case, $\text{Hom}_{\mathcal{C}}(A, B)$ is just a set.

A **simplicial category** is a category enriched in simplicial sets:

$$\mathcal{V} = \text{sSet}.$$

Thus its morphisms are mapping spaces.

A **dg category**, short for **differential graded category**, over a field k is a category enriched in cochain complexes of k -vector spaces:

$$\mathcal{V} = \text{Ch}(k).$$

For a dg category $\mathcal{C}$, its **homotopy category** $H^0(\mathcal{C})$ is the ordinary category with the same objects as $\mathcal{C}$, and with morphism spaces

$$\text{Hom}_{H^0(\mathcal{C})}(A, B) = H^0 \text{Hom}_{\mathcal{C}}^{\bullet}(A, B).$$

A **dg functor** between dg categories is a functor compatible with the dg structures; that is, it sends objects to objects and induces morphisms of complexes

$$\text{Hom}_{\mathcal{C}}^{\bullet}(A, B) \longrightarrow \text{Hom}_{\mathcal{D}}^{\bullet}(F(A), F(B))$$

compatible with composition and units.

A dg functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called a **quasi-equivalence** if the induced morphism of complexes

$$\text{Hom}_{\mathcal{C}}^{\bullet}(A, B) \longrightarrow \text{Hom}_{\mathcal{D}}^{\bullet}(F(A), F(B))$$

is a quasi-isomorphism for all objects $A, B \in \mathcal{C}$, and if the induced functor

$$H^0(F) : H^0(\mathcal{C}) \rightarrow H^0(\mathcal{D})$$

is essentially surjective; i.e., for every object $D \in \mathcal{D}$, there exists an object $C \in \mathcal{C}$ such that $F(C) \cong D$ in $H^0(\mathcal{D})$.

A **right dg module** over $\mathcal{C}$ is a dg functor

$$M : \mathcal{C}^{op} \longrightarrow \text{Ch}(k).$$

Equivalently, it assigns to each object $A \in \mathcal{C}$ a complex $M(A)$, and to each pair of objects $A, B \in \mathcal{C}$ a morphism of complexes

$$M(B) \otimes \text{Hom}_{\mathcal{C}}^{\bullet}(A, B) \longrightarrow M(A),$$

compatible with composition and units. We denote by $\text{Mod}_{dg}(\mathcal{C})$ the dg category of right dg modules over $\mathcal{C}$. For each object $A \in \mathcal{C}$, there is a right dg module

$$h_A := \text{Hom}_{\mathcal{C}}^{\bullet}(-, A).$$

This is called the **right representable dg module** associated to A . Explicitly, it sends an object $X \in \mathcal{C}$ to the complex

$$h_A(X) = \text{Hom}_{\mathcal{C}}^{\bullet}(X, A).$$

A morphism $u : X \rightarrow Y$ acts by precomposition:

$$\mathrm{Hom}_{\mathcal{C}}^{\bullet}(Y, A) \longrightarrow \mathrm{Hom}_{\mathcal{C}}^{\bullet}(X, A), \quad g \longmapsto g \circ u.$$

If $V^{\bullet}$ is a cochain complex, its **shift** $V[1]$ is defined by

$$V[1]^i = V^{i+1}, \quad d_{V[1]}^i = -d_V^{i+1}.$$

If $f : V^{\bullet} \rightarrow W^{\bullet}$ is a morphism of complexes, its **cone** is the complex

$$\mathrm{Cone}(f)^i = W^i \oplus V^{i+1}, \quad d(w, v) = (d_W w + f(v), -d_V v).$$

The dg category $\mathrm{Mod}_{dg}(\mathcal{C})$ has shifts and cones defined objectwise. For example,

$$M[1](A) = M(A)[1],$$

and if $\varphi : M \rightarrow N$ is a closed degree-zero morphism of dg modules, then

$$\mathrm{Cone}(\varphi)(A) = \mathrm{Cone}(M(A) \rightarrow N(A)).$$

A dg category $\mathcal{C}$ is called **pretriangulated** if shifts and cones of representable dg modules are again representable. More precisely, for every $A \in \mathcal{C}$, the shifted module $h_A[1]$ is represented by an object of $\mathcal{C}$, and for every closed degree-zero morphism $f : A \rightarrow B$, the cone of the induced morphism

$$h_A \longrightarrow h_B$$

is also represented by an object of $\mathcal{C}$. In this case, $H^0(\mathcal{C})$ becomes a triangulated category.

Not every dg category is pretriangulated. Let $\mathcal{C}$ be the dg category with one object $*$ and endomorphism

$$\mathrm{Hom}_{\mathcal{C}}(*, *) = k,$$

where k is regarded as a complex concentrated in degree 0. The unique representable right dg module is

$$h_* = \mathrm{Hom}_{\mathcal{C}}(-, *) \simeq k.$$

However, the shifted module

$$h_*[1] \simeq k[1]$$

is not representable by an object of $\mathcal{C}$, since $\mathcal{C}$ has only the object $*$, whose representable module is k , not $k[1]$. Therefore $\mathcal{C}$ is not pretriangulated.

The **dg Yoneda embedding** of dg category $\mathcal{C}$ is the dg functor

$$Y : \mathcal{C} \rightarrow \mathrm{Mod}_{dg}(\mathcal{C}), \quad A \longmapsto \mathrm{Hom}_{\mathcal{C}}^{\bullet}(-, A).$$

The **pretriangulated envelope** of $\mathcal{C}$, denoted $\mathrm{Pretr}(\mathcal{C})$, is the smallest full dg subcategory of $\mathrm{Mod}_{dg}(\mathcal{C})$ which contains all representable dg modules h_A and is closed under finite direct sums, shifts, and cones.

Let $\mathcal{C} = \{*\}$ have the same endomorphism complex k as above. The pretriangulated envelope of $\mathcal{C}$ is obtained by adjoining formal shifts, cones, and finite direct sums:

$$\mathrm{Pretr}(\mathcal{C}) \simeq \mathrm{Perf}(k).$$

Here $\mathrm{Perf}(k)$ denotes the dg category of perfect complexes over k .

Let $\mathcal{C}$ be a pretriangulated dg category. A full dg subcategory $\mathcal{A} \subset \mathcal{C}$ is called **thick** if $H^0(\mathcal{A})$ is a full triangulated subcategory of $H^0(\mathcal{C})$ closed under direct summands. Equivalently, if $X \in H^0(\mathcal{A})$ and Y is a retract of X in $H^0(\mathcal{C})$, then $Y \in H^0(\mathcal{A})$.

Let $\mathcal{S} \subset \text{Ob}(\mathcal{C})$. We denote by

$$\langle \mathcal{S} \rangle$$

the smallest full dg subcategory of $\mathcal{C}$ whose homotopy category is a thick triangulated subcategory of $H^0(\mathcal{C})$ containing $\mathcal{S}$.

Let $\mathcal{C}$ and $\mathcal{D}$ be pretriangulated dg categories. A dg functor

$$F : \mathcal{C} \longrightarrow \mathcal{D}$$

is called **exact** if the induced functor

$$H^0(F) : H^0(\mathcal{C}) \longrightarrow H^0(\mathcal{D})$$

is an exact functor of triangulated categories. Equivalently, it commutes with shifts up to isomorphism and sends distinguished triangles to distinguished triangles.

In practice, we will use the following criterion.

Theorem 3.1 (Generator Criterion for Quasi-Equivalences). Let $F : \mathcal{C} \longrightarrow \mathcal{D}$ be an exact dg functor between pretriangulated dg categories. Let $\mathcal{S} \subset \text{Ob}(\mathcal{C})$ be a collection of objects such that $\langle \mathcal{S} \rangle = \mathcal{C}$ and $\langle F(\mathcal{S}) \rangle = \mathcal{D}$. Suppose that, for all $A, B \in \mathcal{S}$, the induced morphism of complexes

$$\text{Hom}_{\mathcal{C}}^{\bullet}(A, B) \longrightarrow \text{Hom}_{\mathcal{D}}^{\bullet}(F(A), F(B))$$

is a quasi-isomorphism. Then F is a quasi-equivalence.

Therefore, the proof of a dg quasi-equivalence can be reduced to two tasks:

1. Compare the Hom complexes on a set of generators;
2. prove that these generator generate both sides.

3.2 Coherent and Equivariant Coherent Sheaves

Let X be a locally Noetherian scheme. We denote by $\text{Coh}(X)$ the abelian category of coherent $\mathcal{O}_X$ -modules. Let

$$\text{Ch}^b(\text{Coh}(X))$$

be the category of bounded complexes of coherent sheaves and let

$$K^b(\text{Coh}(X))$$

be its homotopy category. It has the same objects as $\text{Ch}^b(\text{Coh}(X))$, but its morphisms are chain maps modulo chain homotopy.

The **bounded derived category** of coherent sheaves is the Verdier localization

$$D^b(\text{Coh}(X)) = K^b(\text{Coh}(X))[\mathcal{W}^{-1}],$$

where $K^b(\text{Coh}(X))$ is the homotopy category of bounded complexes of coherent sheaves and $\mathcal{W}$ is the class of quasi-isomorphisms.

Now suppose that an algebraic group G acts on X . Let

$$a : G \times X \longrightarrow X, \quad p_X : G \times X \longrightarrow X$$

be the action map and the projection to X . Let

$$\mu : G \times G \longrightarrow G$$

be the multiplication map. On $G \times G \times X$, define

$$\begin{aligned} q_X : G \times G \times X &\longrightarrow X, & (g, h, x) &\longmapsto x, \\ q_{23} : G \times G \times X &\longrightarrow G \times X, & (g, h, x) &\longmapsto (h, x), \end{aligned}$$

and

$$\mathrm{id}_G \times a : G \times G \times X \longrightarrow G \times X, \quad (g, h, x) \longmapsto (g, hx).$$

We also regard

$$\mu \times \mathrm{id}_X : G \times G \times X \longrightarrow G \times X, \quad (g, h, x) \longmapsto (gh, x).$$

A *G -equivariant coherent sheaf* on X is a coherent sheaf $\mathcal{F}$ together with an isomorphism

$$\phi : p_X^* \mathcal{F} \xrightarrow{\sim} a^* \mathcal{F}$$

satisfying the following cocycle condition:

$$\begin{array}{ccc} q_X^* \mathcal{F} & \xrightarrow{q_{23}^* \phi} & (a \circ q_{23})^* \mathcal{F} \\ & \searrow (\mu \times \mathrm{id}_X)^* \phi & \downarrow (\mathrm{id}_G \times a)^* \phi \\ & & (a \circ (\mathrm{id}_G \times a))^* \mathcal{F}. \end{array}$$

Here we use the equality

$$a \circ (\mathrm{id}_G \times a) = a \circ (\mu \times \mathrm{id}_X)$$

as maps $G \times G \times X \rightarrow X$.

We denote by

$$\mathrm{Coh}_G(X)$$

the *abelian category of G -equivariant coherent sheaves on X* , and by

$$D^b \mathrm{Coh}_G(X)$$

its *bounded equivariant derived category*.

A complex $\mathcal{E}^\bullet$ of $\mathcal{O}_X$ -modules is called *perfect* if it is locally quasi-isomorphic to a bounded complex of finite rank locally free sheaves. We write

$$\mathrm{Perf}(X)$$

for the *dg category of perfect complexes on X* . If G acts on X , we similarly write

$$\mathrm{Perf}_G(X)$$

for the *dg category of G -equivariant perfect complexes*.

The following standard fact explains why perfect complexes provide the correct dg model for the coherent side of the correspondence. In the smooth case, bounded coherent complexes can be represented by perfect complexes. In the equivariant setting, the same statement holds provided that there are sufficiently many equivariant vector bundles.

Theorem 3.2. Let X be a smooth Noetherian scheme over a field k .

1. The dg category $\mathrm{Perf}(X)$ is a dg enhancement of $D^b(\mathrm{Coh}(X))$. Equivalently,

$$H^0(\mathrm{Perf}(X)) \simeq D^b(\mathrm{Coh}(X)).$$

2. Let G be an algebraic group acting on X . If X has the equivariant resolution property, then $\text{Perf}_G(X)$ is a dg enhancement of $D^b(\text{Coh}_G(X))$. Equivalently,

$$H^0(\text{Perf}_G(X)) \simeq D^b(\text{Coh}_G(X)).$$

Here, We say that X has the **equivariant resolution property** if every G -equivariant coherent sheaf $\mathcal{F} \in \text{Coh}_G(X)$ admits a surjection

$$\mathcal{E} \twoheadrightarrow \mathcal{F},$$

where $\mathcal{E}$ is a G -equivariant vector bundle on X .

3.3 Constructible Sheaves

Let M be a real analytic manifold, and let k be a fixed field of coefficients. We denote by

$$Sh(M)$$

the **dg category of complexes of sheaves of k -vector spaces on M** . Equivalently, $Sh(M)$ may be regarded as a dg enhancement of the derived category of sheaves of k -vector spaces on M .

We first recall the notion of a locally constant sheaf. A sheaf $\mathcal{F}$ of k -vector spaces on M is called **locally constant** if every point $x \in M$ has an open neighborhood U such that

$$\mathcal{F}|_U \simeq k_U^{\oplus r}$$

for some integer r depending on U . A **local system** is a locally constant sheaf whose stalks are finite-dimensional k -vector spaces.

A **stratification** of M is a locally finite decomposition

$$M = \bigsqcup_{\alpha} S_{\alpha}$$

into locally closed smooth submanifolds. We say that A sheaf $\mathcal{F}$ is **constructible** with respect to this stratification if each restriction

$$\mathcal{F}|_{S_{\alpha}}$$

is a local system. We denote by

$$Sh_c(M)$$

the full dg subcategory of $Sh(M)$ consisting of constructible complexes. More precisely, a complex $\mathcal{F}^{\bullet} \in Sh(M)$ is **constructible** if each cohomology sheaf $H^i(\mathcal{F}^{\bullet})$ is constructible and $H^i(\mathcal{F}^{\bullet}) = 0$ for all but finitely many i .

We write

$$Sh_{cc}(M)$$

for the **full dg subcategory of $Sh_c(M)$ consisting of constructible complexes with compact support**. Here the **support of a complex $\mathcal{F}^{\bullet}$** is the union of the supports of its cohomology sheaves.

We write

$$D_S^b(M; k)$$

for the **bounded derived category of k -sheaves constructible with respect to the stratification $\mathcal{S} = \{S_{\alpha}\}_{\alpha \in A}$** .

Let

$$j : U \hookrightarrow M$$

be the inclusion of a locally closed subset. The **standard sheaf associated to U** is

$$j_! k_U,$$

where $j_!$ denotes extension by zero. The **costandard sheaf associated to U** is

$$j_* k_U.$$

Here k_U denotes the constant sheaf on U with coefficients in the field k .

These sheaves are basic building blocks of constructible categories. In many situations, constructible sheaves are generated by standard or costandard sheaves associated to the strata of a chosen stratification.

Theorem 3.3 (Generation by standard and costandard sheaves). Let M be a real analytic manifold, and let

$$M = \bigsqcup_{\alpha \in A} S_\alpha$$

be a finite stratification by locally closed submanifolds. Let $j_\alpha : S_\alpha \hookrightarrow M$ denote the inclusion. Then

$$D_S^b(M; k) = \langle j_{\alpha!} L \mid \alpha \in A, L \text{ a finite-dimensional local system on } S_\alpha \rangle.$$

Equivalently,

$$D_S^b(M; k) = \langle j_{\alpha*} L \mid \alpha \in A, L \text{ a finite-dimensional local system on } S_\alpha \rangle.$$

In particular, if every stratum S_α is contractible, then

$$D_S^b(M; k) = \langle j_{\alpha!} k_{S_\alpha} \mid \alpha \in A \rangle = \langle j_{\alpha*} k_{S_\alpha} \mid \alpha \in A \rangle.$$

3.4 Singular Support

Let M be a real analytic manifold, and let T^*M denote its cotangent bundle. The cotangent bundle carries a natural action of $\mathbb{R}_{>0}$ by dilation along the fibers:

$$t \cdot (x, \xi) = (x, t\xi), \quad t \in \mathbb{R}_{>0}.$$

A subset $\Lambda \subset T^*M$ is called **conic** if it is invariant under this action; equivalently,

$$(x, \xi) \in \Lambda \implies (x, t\xi) \in \Lambda \quad \text{for all } t \in \mathbb{R}_{>0}.$$

For a constructible complex of sheaves $\mathcal{F} \in Sh_c(M)$, its **singular support**, also called its **microsupport**, is a closed conic subset

$$SS(\mathcal{F}) \subset T^*M.$$

The singular support is a microlocal refinement of the ordinary support. A covector $(x_0, \xi_0) \in T^*M$ is not contained in $SS(\mathcal{F})$ if there exists a conic open neighborhood $U \subset T^*M$ of (x_0, ξ_0) such that, for every C^1 -function φ defined near x_0 satisfying

$$\varphi(x_0) = 0, \quad (x_0, d\varphi_{x_0}) \in U,$$

one has

$$(R\Gamma_{\{\varphi \geq 0\}}(\mathcal{F}))_{x_0} = 0.$$

Here $R\Gamma_Z(\mathcal{F})$ denotes the derived functor of sections with support in a closed subset $Z \subset M$. Thus

$$(R\Gamma_{\{\varphi \geq 0\}}(\mathcal{F}))_{x_0}$$

is the local cohomology complex of $\mathcal{F}$ at x_0 with support in the half-space $\{\varphi \geq 0\}$.

Equivalently, $SS(\mathcal{F})$ is the closed conic subset of T^*M consisting of covectors along which such local cohomological vanishing fails. It records the cotangent directions in which sections of $\mathcal{F}$ cannot be propagated microlocally.

If $\mathcal{F}$ is locally constant on M , then it has no nonzero singular directions. Hence

$$SS(\mathcal{F}) \subset T_M^*M,$$

where $T_M^*M \subset T^*M$ denotes the zero section.

Let $S \subset M$ be a locally closed submanifold. The **conormal bundle of S in M** is

$$T_S^*M = \{(x, \xi) \in T^*M \mid x \in S, \xi|_{T_x S} = 0\}.$$

If $\mathcal{F}$ is constructible with respect to a stratification

$$M = \bigsqcup_{\alpha \in A} S_\alpha,$$

then its singular support is contained in the union of the conormal bundles to the strata:

$$SS(\mathcal{F}) \subset \bigcup_{\alpha \in A} T_{S_\alpha}^*M.$$

This gives the basic geometric control on the singularities of a constructible sheaf.

Suppose that $\Lambda \subset T^*M$ be a closed conic subset. We define

$$Sh_\Lambda(M) := \{\mathcal{F} \in Sh_c(M) \mid SS(\mathcal{F}) \subset \Lambda\}.$$

to be the **full dg subcategory of $Sh_c(M)$ consisting of constructible complexes whose singular support is contained in Λ** .

Thus Λ imposes a microlocal constraint on constructible sheaves. Instead of allowing arbitrary constructible singularities, one only allows singular behavior in the cotangent directions prescribed by Λ .

Theorem 3.4. Let $\Lambda \subset T^*M$ be a closed conic subset. The full subcategory

$$Sh_\Lambda(M) \subset Sh_c(M)$$

is closed under shifts, cones, finite direct sums, and retracts. Consequently, its homotopy category

$$H^0(Sh_\Lambda(M))$$

is a triangulated subcategory of $H^0(Sh_c(M))$.